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November 23, 2022

Attn: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

10 CFR 50.73

**SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-388/2022-001-00
UNIT 2 LICENSE NO. NPF-22
PLA-8033**

Docket No. 50-388

Attached is Licensee Event Report (LER) 50-388/2022-001-00. The LER reports an event in which both divisions of the Core Spray system were concurrently inoperable. This condition is being reported in accordance with 10 CFR 50.73(a)(2)(v)(D) as an event or condition that could have prevented fulfillment of a safety function.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new or revised regulatory commitments.

 **DLAMARCA FOR K CIMORELLI**
K. Cimorelli

Attachment: LER 50-388/2022-001-00

Copy: NRC Region I
Mr. C. Highley, NRC Senior Resident Inspector
Ms. A. Klett, NRC Project Manager
Mr. M. Shields, PA DEP/BRP

**LICENSEE EVENT REPORT (LER)**

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form <https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk a/c: aira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Susquehanna Steam Electric Station, Unit 2						2. Docket Number 05000388			3. Page 1 of 3		
4. Title Inadequate Performance of Loss of Safety Determination Resulting in Both Divisions of Core Spray Being Inoperable											
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number	
09	26	2022	2022	- 001 -	00	11	23	2022	Facility Name	Docket Number	
									05000		
									05000		
9. Operating Mode 1						10. Power Level 100					
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)											
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)			
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73			
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(4)			
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.71(a)(5)			
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.73(a)(2)(i)(B)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(1)(i)			
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(i)			
☐ 20.2203(a)(2)(iii)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.77(a)(2)(ii)			
☐ 20.2203(a)(2)(iv)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)					
☐ 20.2203(a)(2)(v)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)					
☐ Other (Specify here, in Abstract, or in NRC 366A).											
12. Licensee Contact for this LER											
Licensee Contact Derek R. Smith, Senior Engineer – Nuclear Regulatory Affairs									Phone Number (Include Area Code) (570) 542-1377		
13. Complete One Line for each Component Failure Described in this Report											
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS		
14. Supplemental Report Expected									15. Expected Submission Date		
☒ No		☐ Yes (If yes, complete 15. Expected Submission Date)							Month	Day	Year
16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)											
On September 26, 2022, operators at the Susquehanna Steam Electric Station performed quarterly stroke time testing of Unit 2 Emergency Service Water (ESW) to Turbine Building Closed Cooling Water and ESW to Reactor Building Closed Cooling Water isolation valves. Performance of this surveillance results in the inoperability of Unit 2, Division 2 of the Core Spray system. This surveillance was performed while Unit 2, Division 1 of the Core Spray system was out of service for planned maintenance, resulting in both divisions of the Core Spray system being inoperable and a loss of safety function. The condition was identified during the subsequent shift while performing routine review of operator logs. The condition is being reported in accordance with 10 CFR 50.73(a)(2)(v)(D) as a condition that could have prevented fulfillment of a safety function. The cause of the event was determined to be inadequate performance of a loss of safety function determination, which failed to identify a loss of safety function resulting from concurrent inoperability of both divisions of Core Spray. This cause was attributed to human performance. There were no actual consequences to the health and safety of the public as a result of this event.											



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Susquehanna Steam Electric Station, Unit 2	05000-388	YEAR	SEQUENTIAL NUMBER	REV NO.
		2022	- 001 -	00

NARRATIVE

CONDITIONS PRIOR TO EVENT

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power (RTP)

Unit 2 – Mode 1, approximately 100 percent RTP

EVENT DESCRIPTION

On September 26, 2022, operators at the Susquehanna Steam Electric Station performed quarterly stroke time testing of the Unit 2 Emergency Service Water (ESW) [EIS System Code: BI] to Turbine Building Closed Cooling Water (TBCCW) [EIS System Code: CC] and ESW to Reactor Building Closed Cooling Water (RBCCW) [EIS System Code: CC] isolation valves [EIS Component Code: ISV]. Performance of this surveillance results in the inoperability of Unit 2, Division 2 of the Core Spray system [EIS System Code: BG]. This surveillance was performed while Unit 2, Division 1 of the Core Spray system was out of service for planned maintenance, resulting in both divisions of the Core Spray system being inoperable and a loss of safety function. The condition was identified during the subsequent shift while performing routine review of operator logs.

The following timeline demonstrates the sequence of events:

09/25/2022 ~ 09:00: The quarterly surveillance original start time is postponed due to conflicting work on the Unit 2 RBCCW heat exchangers [EIS Component Code: HX]. A condition report is generated for the surveillance to be rescheduled after the completion of the Unit 2, Division 1 Core Spray planned maintenance.

09/25/2022 ~ 19:45: Unit 2, Division 1 Core Spray planned maintenance begins. Limited Condition of Operation (LCO) 3.5.1, "Emergency Core Cooling Systems (ECCS), Reactor Pressure Vessel (RPV) Water Inventory Control, and Reactor Core Isolation Cooling (RCIC) System," Condition A is entered to support this work.

09/26/2022 ~ 05:00: During operations shift turnover, the need to track the postponed surveillance was discussed. However, the oncoming Unit Supervisor completed turnover with the misunderstanding that the surveillance needed to be completed during the Unit 2 RBCCW heat exchanger restoration.

09/26/2022 ~ 09:00: The surveillance is rescheduled for 09/27/2022, after the Unit 2, Division 1 Core Spray planned maintenance was scheduled to be complete.

09/26/2022 ~ 14:00: Unit 2 RBCCW heat exchanger work is reported completed. Operations begins preparation for performance of the quarterly surveillance. During this process, operations inadequately performed a loss of safety function determination using a previously completed copy of the station procedure, which did not identify the Unit 2, Division 1 Core Spray ongoing planned maintenance.

09/26/2022 ~ 14:08: Operations entered Condition C of LCO 3.7.2, "Emergency Service Water (ESW) System," due to Division 2 ESW being inoperable for performance of the surveillance. A peer check for entry into the LCO condition was not received as required by station procedures. Investigation identified this as the time the loss of safety function occurred.

09/26/2022 ~ 14:17: The surveillance on Division 2 ESW is completed. Operations exited Condition C of LCO 3.7.2.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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Susquehanna Steam Electric Station, Unit 2	05000-388	2022	- 001 -	00

NARRATIVE

09/26/2022 ~ 19:00: The subsequent Operating crew on the next shift identified the loss of safety function during routine review of operator logs; this review determined the condition existed for approximately 9 minutes. The condition was entered into the corrective action program.

This condition is being reported in accordance with 10 CFR 50.73(a)(2)(v)(D) as a condition that could have prevented fulfillment of a safety function.

CAUSE OF EVENT

The cause of the event was determined to be inadequate performance of a loss of safety function determination by the Operator. The review failed to identify a loss of safety function resulting from concurrent inoperability of both divisions of Core Spray, Division 1 of Core Spray for planned maintenance and Division 2 of Core Spray due to performance of a surveillance. This cause was attributed to human performance.

ANALYSIS/SAFETY SIGNIFICANCE

Based on engineering analysis of the event, Unit 2, Division 2 Core Spray could have performed its safety function for this condition. The Unit 2, Division 2 core spray room temperatures were evaluated with two core spray pumps in operation without room coolers. Assuming the worst case, bounding scenario of complete rupture of the TBCCW/RBCCW piping with the ESW isolation valves open, the resulting Core Spray room temperatures would not have exceeded functionality limits. Therefore, Core Spray could have fulfilled its safety function.

This event will not be counted as a safety system functional failure for the NRC performance indicator based on the engineering analysis supporting the system's ability to fulfill the safety function. There were no actual consequences to the health and safety of the public as a result of this event.

CORRECTIVE ACTIONS

1. Coaching was performed with applicable operators per the station accountability and engagement process procedure.
2. Revoked licensed operator qualifications for applicable operators pending remediation.
3. Perform focused observations of performance of loss of safety function determinations to assess Senior Reactor Operators proficiency levels in the task.

COMPONENT FAILURE INFORMATION

Not applicable.

PREVIOUS OCCURRENCES

None.